Evolution and Reinforcement Learning in Changing Environments

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Abstract

We compare ecological neuroevolution (OE), evolution strategies (ES) and proximal policy optimization (PPO) in a simulated foraging world. Fixed- and varied-environment configurations are tuned separately and trained for the same nominal agent-step budget, with ten runs per configuration. We evaluate food acquisition before and during adaptation to thirty test environments, and population persistence in a separate assay. On the twenty shifted environments, OE scores below PPO with domain randomization: the difference in interquartile mean of normalized adaptation area under the curve is -0.5592 (nominal 95% bootstrap interval $[-0.6088, -0.5196]$). Varied OE also scores below OE-fixed. ES-DR scores slightly above ES-fixed, while the PPO comparison does not support the predicted benefit of varied training. OE depends heavily on replacement agents supplied by a population floor; these account for over 99% of births during resource-scarce adaptation. Without floor support, OE has poorer population persistence than PPO-DR. The comparisons include differences in selected hyperparameters, objectives and population structure.

1 Introduction

Milano et al. (2019) found that environmental variation across generations improved evolved controllers, with the best performance at a moderate rate of change. Ecological neuroevolution imposes a different selection pressure: agents must acquire enough food to survive and reproduce. Food acquisition also determines task performance, but selection through survival may favor different controllers from direct optimization of food intake.

We compare ecological neuroevolution (OE), evolution strategies (ES) and proximal policy optimization (PPO) in a continuous two-dimensional foraging world. OE maintains an interacting population with energy-dependent death and reproduction, supplemented by a minimum population of externally supplied replacements. ES and PPO optimize food acquisition and deploy copies of a single controller. Each method has separately tuned fixed- and varied-environment configurations. We evaluate food acquisition before and during adaptation, then measure population persistence in a separate assay without replacements.

A supporting experiment with inherited traits and fixed behavioral rules examines search objectives, mutation and cloning (Appendix D).

We predicted that varied-environment OE would outperform PPO with domain randomization during adaptation. OE instead starts below PPO-DR and remains below it throughout the measured interval. Varied OE also scores below fixed OE; a later ES comparison favors the varied configuration by a smaller amount.

2 Related work

Milano et al. (2019) varied the rate of environmental change during evolution. Milano and Nolfi (2024) compared ES and PPO across tasks and treatments of environmental variation. Our experiment compares separately selected fixed- and varied-training configurations in a foraging task without sweeping the rate of environmental change.

Hamon et al. (2023) studied persistent neural-agent populations with physiology-dependent reproduction and death. Nisioti and Moulin-Frier (2022) used a minimal ecological model and ablations to distinguish fitness-based selection from niche-limited competition. Our implementation also couples survival and reproduction to agent behavior, but uses feedforward controllers and externally supplied population replacements.

Prior comparisons examine curricula (Nisioti et al., 2025), sequential changes and forgetting (Nisioti et al., 2026), and update and inheritance regimes in ecological evaluations (Ge and Cheng, 2026). Here, adaptation begins independently from the trained snapshot in each test environment and permits further parameter updates.

3 Experimental design

3.1 Foraging environment and controllers

Agents move in a square arena with walls and regenerating food. Their observations contain 30 values: eight directional sectors for each of food, other agents and walls; current energy; the previous two-dimensional action; and three body traits. A feedforward controller has two hidden layers of 32 tanh units and two outputs, controlling turn and forward speed. Every method uses this actor architecture. Maximum speed, size and sensing range are fixed at 1, 1 and 10, respectively.

Energy decreases with time, movement and sensing. A food item supplies 40 energy units; an agent starts with 100 and dies when its energy reaches zero. In ecological mode, agents can reproduce above an energy threshold, paying an energy cost for each offspring; they also die at a maximum age. Appendix A gives the full settings. Episodic competence evaluation uses the same movement, sensing and food dynamics, without reproduction.

Varied training samples arena size, food supply, layout and regrowth from the ranges in Table 1. The fixed environment has side length 60, 40 food items, a uniform layout and regrowth probability 0.03. Validation uses 20 sampled training-range environments and four corner cases. Test conditions are fixed in advance of final evaluation: ten training-range environments (T1), ten smaller and resource-scarce clustered environments (T2), and ten environments with food drift and seasonal relocation (T3). T3 retains the training ranges for the remaining parameters. T2 tests resource scarcity; T3 tests changing food locations.

3.2 Training procedures

Ecological neuroevolution. OE starts with 32 agents and permits at most 128. Each agent has its own controller. Offspring inherit the parent's weights with Gaussian perturbations; lineage survival depends on death, reproduction and the population-floor rule described below. In varied OE, the environmental parameters change every 2,000 world steps while the population continues. OE-fixed retains the canonical environment. During adaptation, either variant continues evolving in the assigned test environment.

Table 1: Training ranges and test suites. Each test suite contains ten environments; layout parameters apply only to the corresponding layout.

Parameter	Varied training / T1	T2	T3
Arena side L	40–80	30–40	40–80
Food items F	20–60	10–20	20–60
Layouts	Uniform, clusters, patches	Clusters	Uniform, clusters, patches
Cluster count k	2–5	2–3	2–5
Cluster scale σ	3–8	2–4	3–8
Patch radius	4–10	Not used	4–10
Regrowth probability	0.01–0.05	0.005–0.01	0.01–0.05
Food drift	0	0	0.05–0.3
Season period (steps)	None	None	100–300

Both variants include a population floor of 16. After a step, vacant slots are replenished with mutated offspring of uniformly sampled surviving, non-newborn agents; if that pool is empty, agents alive at the start of the step supply the parent pool. Replacements receive birth energy and a new position. Replacements can maintain the population when natural reproduction is insufficient. We count natural and floor births separately.

Evolution strategies. ES optimizes the shared actor weights using the `Open_ES` implementation in evosax (Salimans et al., 2017; Lange, 2023). It uses antithetic perturbations and centered-rank fitness shaping, with candidate fitness determined by episodic food acquisition. ES-DR samples training environments; ES-fixed uses the canonical environment. Evaluation uses the strategy mean.

Each candidate is evaluated with 16 policy copies in 500-step worlds without births. Its score is food per initial agent, averaged over its configured episodes. Candidates in a generation share environment draws and reset/step random keys; ES-DR redraws the episode environments each generation.

Proximal policy optimization. PPO uses a clipped policy-gradient objective (Schulman et al., 2017), a separate value network, and a parameter-shared policy across agents in 64 parallel episodic worlds. Food acquisition supplies the reward. PPO-DR samples environment parameters at episode resets; PPO-fixed uses the canonical environment. Training actions are stochastic, whereas competence evaluation uses the actor's deterministic output. The implementation adapts PureJaxRL's continuous-action PPO (Appendix A).

Each world contains 16 agents and restarts at 500 steps or when all agents die. An agent's death terminates its trajectory. Steps beginning with a dead agent are excluded from losses and advantage normalization; when enabled, reward-normalization statistics are collected over living agents.

OE maintains a heterogeneous population; ES and PPO deploy copies of one policy. Table 2 summarizes their deployment and transfer operations.

3.3 Tuning, budgets and evaluation

Eight random-search trials were run separately for each variant, each with a distinct tuning seed and 5×10^7 nominal living-agent steps. The selection score was mean zero-shot food acquisition

Table 2: Training and transfer procedures. Each row has fixed and varied variants, separately tuned. Body traits and actor architecture are shared.

Procedure	Training signal	Competence deployment	State at adaptation start
OE	Energy-dependent survival/reproduction plus floor replacements	16 sampled population members	Genomes/alive mask retained; world state and energy reset
ES	Rank-shaped episodic food score	16 copies of mean actor	Mean solution retained; fresh strategy/optimizer
PPO	Food reward and clipped policy objective	16 copies of actor	Actor/critic and reward statistics retained; fresh Adam/rollout state

above the random-policy score across the 24 validation environments. Tuning therefore used a different endpoint from the primary adaptation AUC. The selected configuration was then trained with ten final seeds, 5000–5009, using 2×10^8 nominal living-agent steps per run. All ten final runs per variant are retained. Settings and search ranges appear in Appendix A.

One living-agent step counts an agent alive at the beginning of a simulated step. Wall time, floating-point operations, optimizer updates and checkpoint positions can differ at the same nominal agent-step budget. Updates finish in indivisible units and may cross a stopping target. We analyze both the prespecified nominal grid and the actual step counts.

Competence is total food consumed divided by the 16 initial agents in a 500-step episode, averaged across eight evaluation episodes. OE samples population members afresh for each episode, without replacement when at least 16 are alive; ES and PPO use 16 policy copies. Episode and reset seeds are shared across methods, although the random-action reference does not share every subsequent world-step random draw with learned policies.

Adaptation starts independently from the final trained snapshot in each test environment and receives 2×10^7 nominal living-agent steps. There are eleven competence measurements: the zero-shot score and ten subsequent checkpoints. OE retains genomes and the alive mask while resetting physical state and energy. PPO retains actor/critic parameters and reward-normalization statistics but resets Adam and rollout state. ES retains its mean solution with fresh optimizer and strategy state.

Persistence is evaluated separately, before adaptation. Thirty-two sampled OE agents or policy copies enter ecological mode for 5,000 world steps, with reproduction but no mutation and no population floor. We record extinction and steps survived.

3.4 Outcomes and statistical analysis

Let $c_{mre}(t)$ be competence for method m, run r and environment e at normalized adaptation checkpoint $t \in \{0, 0.1, \ldots, 1\}$. We use the prespecified transformation

$$z_{mre}(t) = \frac{c_{mre}(t) - q_e}{b_e - q_e}, \qquad A_{mre} = \frac{1}{10} \sum_{j=1}^{10} \frac{z_{mre}(t_{j-1}) + z_{mre}(t_j)}{2}, \tag{1}$$

where q_e is the random-policy score and b_e is the largest observed score across the five core methods, all final runs and all eleven checkpoints in that environment. No environment had $b_e \leq q_e$, so none was excluded. The later ES-fixed analysis uses the saved core anchors.

The primary effect is

$$D = \mathrm{IQM}\{A_{\mathrm{OE},r,e} : e \in T2 \cup T3\} - \mathrm{IQM}\{A_{\text{PPO-DR},r,e} : e \in T2 \cup T3\}, \tag{2}$$

where the interquartile mean (IQM) averages the middle half of the pooled run–environment values. We bootstrap complete training runs, keeping their environment outcomes together, with 50,000 draws and statistics seed 4800 (Agarwal et al., 2021; Henderson et al., 2018). The prespecified margin is 0.05 on this reference scale. AUC measures performance over the adaptation interval, including starting competence.

The seven core secondary comparisons use two-sided Mann–Whitney tests on per-run environment means, with Holm adjustment within the family. The later ES comparisons form a separate family. These rank tests and the pooled-IQM effects summarize different aspects of performance. Appendix B gives all comparisons and decision rules.

Neural intervals are nominal percentile bootstrap intervals over training runs, holding the normalization anchors and test environments fixed. They exclude uncertainty from anchor selection, environment and episode sampling, and tuning; selection of the maximum anchors from the same runs leaves their coverage uncertain. Appendix C examines anchor re-estimation and paired resampling of corresponding seeds.

4 Results

4.1 OE scores below PPO-DR during adaptation

OE performs below PPO-DR, contrary to the primary prediction. Across T2 and T3, OE has a pooled AUC IQM of 0.2905, compared with 0.8497 for PPO-DR. Their difference is -0.5592 with a nominal 95% interval of $[-0.6088, -0.5196]$, entirely below the negative margin. OE also scores below ES-DR in the prespecified S1 comparison: the IQM difference is -0.5871 and the Holm-adjusted rank-test p value is 0.00110. Figure 1 shows estimates for all six procedures.

OE is already below PPO-DR before adaptation. The zero-shot IQM difference on T2/T3 is -0.4648; at the final checkpoint it is -0.6035 (S4 and S5, both Holm-adjusted $p = 0.00110$). Figure 3 shows the mean trajectories. In T2, OE's mean normalized competence declines from 0.4383 to 0.2338, while PPO-DR changes from 0.7244 to 0.7478. T3 shows a smaller decline for OE, from 0.2966 to 0.2655.

On the in-range T1 suite, the OE–PPO-DR difference in adaptation AUC is -0.5371 (nominal 95% interval $[-0.5951, -0.4799]$; S6).

4.2 Fixed and varied training

The separately tuned varied OE procedure scores below OE-fixed on T2/T3 AUC: difference -0.1210, interval $[-0.1799, -0.0680]$ (S2; Holm-adjusted $p = 0.00440$). For PPO, the corresponding difference is 0.0099, interval $[-0.0004, 0.0201]$. Its predicted direction is unsupported under the prespecified rank-test rule (S3; adjusted $p = 0.6232$).

In the later ES-fixed extension, ES-DR has higher T2/T3 adaptation AUC by 0.0286 (nominal 95% interval $[0.0203, 0.0384]$), below the 0.05 margin. The corresponding rank test gives adjusted $p = 0.000365$. ES-DR also has higher T1 zero-shot competence by 0.0645 (interval $[0.0499, 0.0788]$; adjusted $p = 0.000365$).

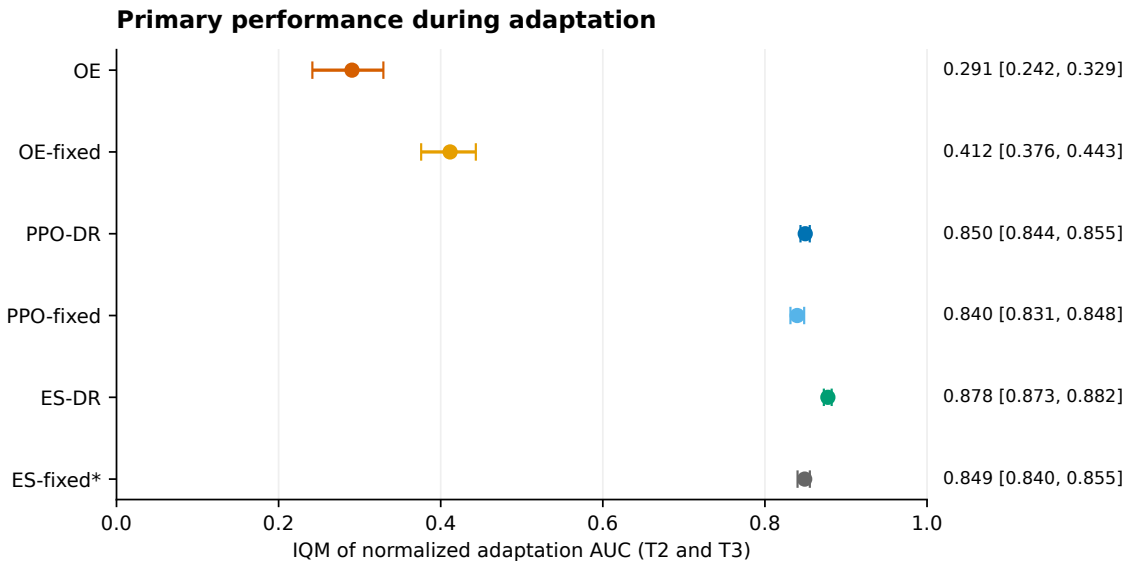


Figure 1: Performance during adaptation on T2 and T3. Points show pooled IQMs of normalized AUC; bars are nominal 95% whole-run percentile bootstrap intervals from 50,000 draws. Each procedure has ten training runs evaluated on twenty fixed environments. Asterisk: ES-fixed was added after the core experiment.

Table 3: Extinction frequencies in the 5,000-step persistence assay without floor support. Each cell pools ten training runs evaluated in ten environments. Asterisk: ES-fixed extension.

Procedure	T1	T2	T3
OE	0.48	0.94	0.76
OE-fixed	0.30	0.93	0.48
PPO-DR	0.03	0.83	0.15
PPO-fixed	0.10	0.95	0.14
ES-DR	0.03	0.89	0.22
ES-fixed*	0.11	0.87	0.21

Selected hyperparameters also differ between fixed and varied configurations. For OE, the mutation standard deviation is 0.03412 under varied training and 0.00507 under fixed training.

4.3 Population persistence and support

OE has higher extinction in the separate persistence assay than PPO-DR. Across T2/T3, the S7 difference in IQMs of per-run extinction fractions is 0.3750, interval $[0.2500, 0.4833]$, against the predicted OE advantage (adjusted $p = 0.00110$). Table 3 reports the descriptive extinction frequencies for every procedure. Extinction is frequent across all six procedures in resource-scarce T2, including those with high normalized competence. High relative food acquisition can coexist with insufficient intake for population survival.

The population floor makes a substantial contribution during OE training and adaptation. The mean run-level share of training births supplied by the floor is approximately 58% for OE and 59% for OE-fixed. During T2 adaptation, the pooled shares rise to 99.53% and 99.60%, respectively

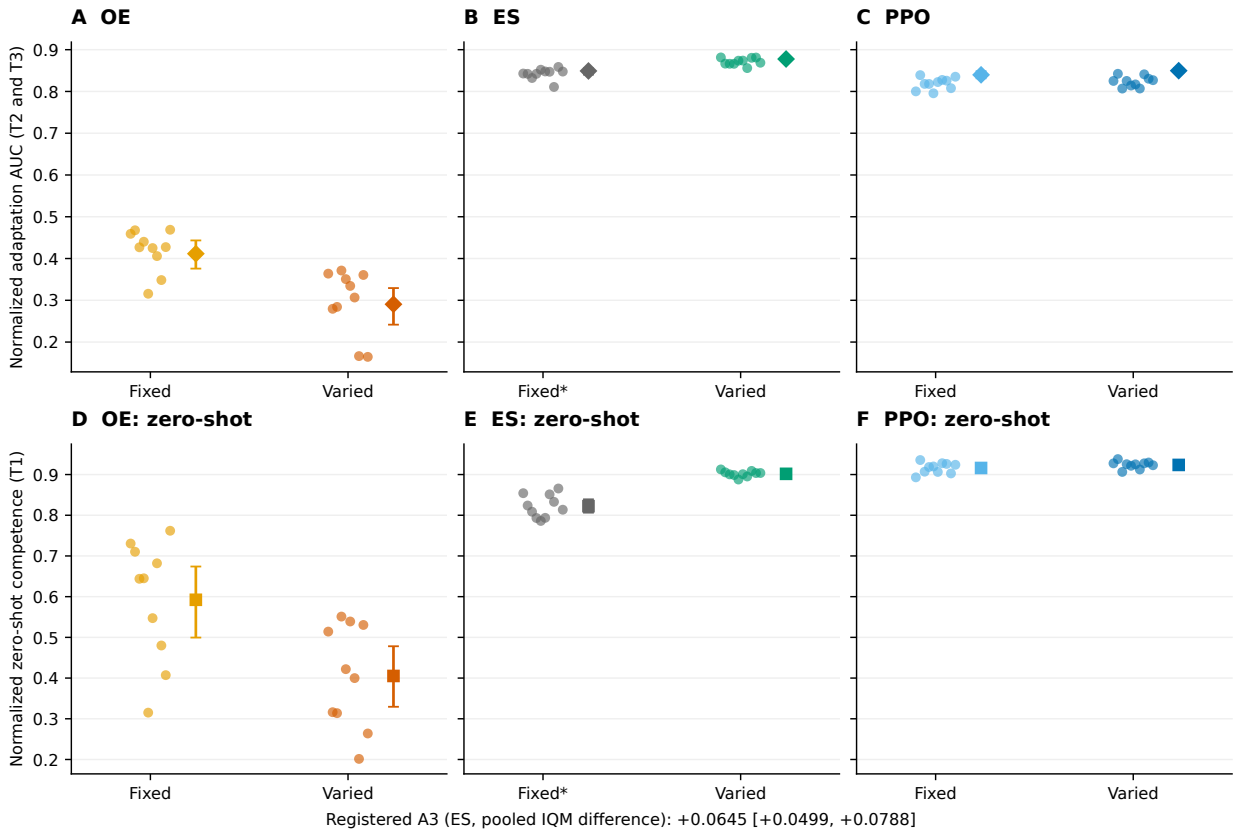


Figure 2: Fixed and varied configurations. Upper panels: T2/T3 AUC, with run means (circles), pooled IQMs (diamonds) and nominal 95% whole-run intervals. Lower panels: T1 zero-shot run means and their mean (squares), with descriptive 95% intervals from 20,000 whole-run bootstrap draws. A3 denotes the pooled-IQM contrast; lower OE/PPO panels are descriptive. Each configuration was tuned separately. Asterisk: ES-fixed extension.

(Table 4).

4.4 Learning curves

Between the first validation checkpoint near 20 million steps and the end of training at 200 million steps, OE's mean score above random increases from 9.4818 to 10.6102; six of ten runs improve (Figure 3). Three additional exploratory runs extend OE training to one billion steps (Appendix C).

The OE–PPO-DR contrast remains negative under paired-seed resampling, re-estimation of normalization anchors and integration over actual step counts. The weaker PPO fixed/varied comparison depends on the summary and resampling method (Appendix C).

Table 4: Share of new agents supplied by population-floor replacements during adaptation. Fractions pool event counts over ten runs and ten environments per suite.

Suite	OE	OE-fixed
T1	64.35%	59.29%
T2	99.53%	99.60%
T3	78.90%	71.87%

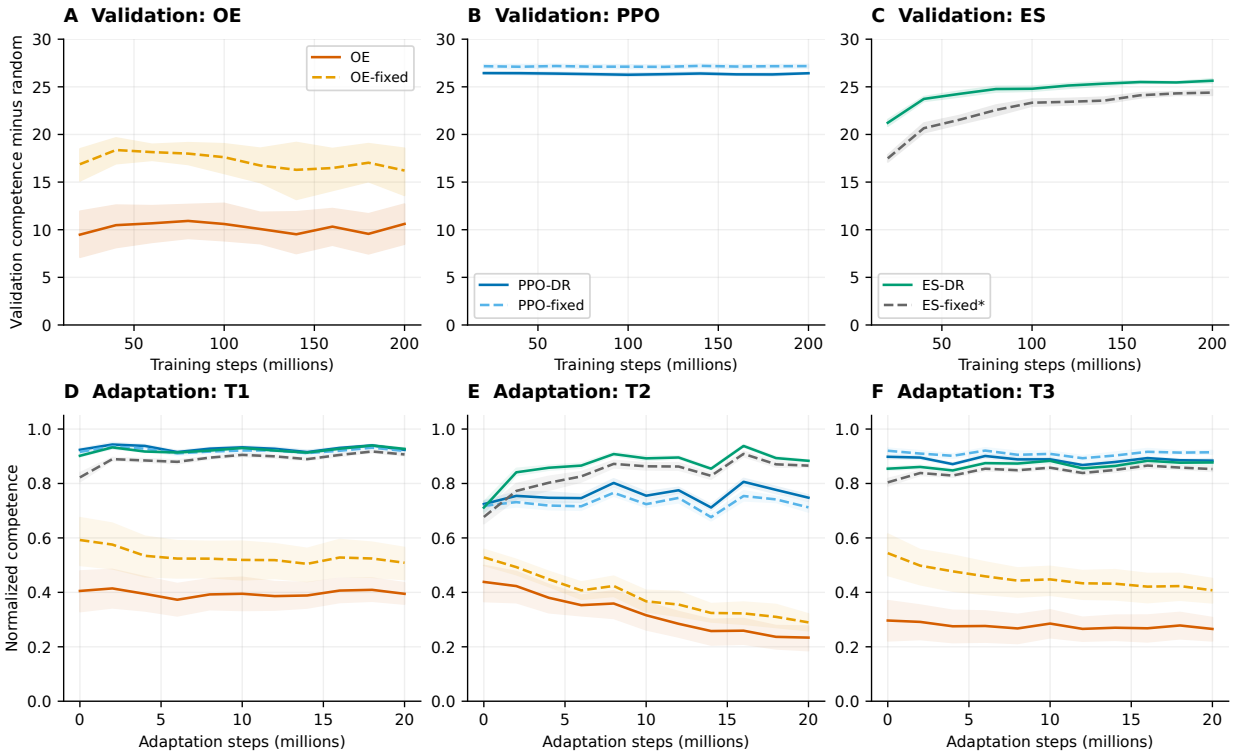


Figure 3: Training and adaptation. Upper panels: validation competence above the random reference. Lower panels: normalized test competence during adaptation. Lines average ten runs; shading gives pointwise 95% whole-run percentile intervals from 20,000 draws. Axes use nominal living-agent steps. Training measurements begin near 20 million steps.

5 Discussion

OE scores below PPO-DR before and during adaptation, and its populations have poorer persistence without floor support. The methods differ in several respects: OE selects an interacting population through survival and reproduction, whereas ES and PPO optimize episodic food acquisition and deploy copies of one actor. Separate tuning adds differences in hyperparameters. The experiment compares these configurations without isolating the contribution of each component.

Varied training has different outcomes across methods. The positive ES zero-shot contrast on T1 agrees in direction with Milano et al. (2019); its adaptation contrast on the shifted suites is smaller. Varied OE scores below OE-fixed. We did not sweep the rate of environmental change, so the comparison does not test the moderate-rate optimum found in their study.

In resource-scarce T2, even policies with high normalized food scores frequently go extinct.

Relative food acquisition and population survival measure different outcomes: survival requires enough intake to cover energy costs. Floor replacements supply additional energy during OE training and adaptation and account for over 99% of births in T2. This dependence makes the population floor a central part of the tested OE procedure.

The results cover one foraging world and thirty fixed test environments. Tests in other environments and comparisons with matched hyperparameters, objectives and population structure would help determine which aspects of the procedures account for the observed differences.

6 Conclusion

In this foraging task, ecological neuroevolution performs below PPO-DR during adaptation and has poorer persistence without replacement support. Varied training scores below fixed training for OE, slightly above it for ES, and gives no supported advantage for PPO under the prespecified test.

Data and code availability

Code, data and manuscript source are available at `https://github.com/Guntikiwritescode/ecological-neuroevolution`. Reproduction instructions are in `REPRODUCE.md`.

Author contributions

Abhinav Guntaka designed the study, implemented the software, ran the experiments, analyzed the results and wrote the manuscript.

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A Implementation and selected settings

The actor has dimensions $30 \to 32 \to 32 \to 2$, with tanh hidden activations and 2,114 parameters. Initial weights are orthogonal, with gain $\sqrt{2}$ in hidden layers and 0.01 in the action layer; biases start at zero. The world maps the outputs through tanh to a turn bounded by $\pi/4$ per step and a speed between zero and the body's maximum. Evaluation actions are deterministic. The observation includes clipped energy divided by birth energy and scaled body traits, even though the latter remain fixed.

The per-step energy cost is $0.5 + s^3v^2 + 0.05r$, where s is size, v is commanded speed after action squashing but before position clipping at a wall, and r is sensing range. With the default body it is $1 + v^2$. Food supplies 40 energy units. Ecological reproduction requires energy at least 200, costs the parent 120 and supplies the offspring with 100. Maximum age is 1,000 steps and the offspring placement offset is 1. The ecological capacity is 128; competence evaluation has 16 agent slots. The floor rule described in Section 3.2 adds energy from outside this budget and is logged separately.

Within a step, agents move, consume reachable food, pay costs, die and reproduce; food dynamics then update. Random priority resolves contested food. Births require free slots, with eligible parents processed in random order, at most one child per parent per step, and payment only for successful births. At a varied-OE environment change, positions are clipped to the new arena as needed, while energy, age and genomes persist. Food is reinitialized under the new parameters.

For PPO, the common settings are 64 parallel worlds, 32 minibatches, GAE parameter 0.95, clipping parameter 0.2, value-loss coefficient 0.5, maximum gradient norm 0.5, initial log standard deviation zero and no learning-rate annealing. The separately selected values are in Table 5. PPO's actor parameters, critic parameters, log standard deviation and reward-normalization state are carried into adaptation; its return accumulator, optimizer, environment state, counters and random stream are reset. OE carries controller genomes and the alive mask, while resetting positions, headings, energy, age, actions and counters. ES carries its mean solution while restarting its optimizer/strategy and generation counter.

The PPO critic has the same two hidden layers as the actor and a scalar output. Action exploration is Gaussian with learned, state-independent log standard deviation. The adapted implementation starts from PureJaxRL commit `31756b1`; ES uses evosax version 0.3.2.

Table 5: Selected hyperparameters, rounded to six significant digits. Exact values are in the accompanying tuning CSV. Asterisk: ES-fixed extension.

Procedure	Settings
OE	Mutation SD = 0.0341186
OE-fixed	Mutation SD = 0.00507475
PPO-DR	Entropy coefficient = 0; Discount = 0.99; Learning rate = 0.000180353; Reward normalization = on; Rollout length = 64; Update epochs = 8
PPO-fixed	Entropy coefficient = 0; Discount = 0.95; Learning rate = 0.000706297; Reward normalization = off; Rollout length = 128; Update epochs = 8
ES-DR	Episodes per candidate = 1; Learning rate = 0.0073746; Population size = 64; Perturbation SD = 0.089879
ES-fixed*	Episodes per candidate = 4; Learning rate = 0.0117805; Population size = 32; Perturbation SD = 0.0441052

The tuning search for PPO sampled learning rate log-uniformly from 10^{-4} to 10^{-3}, rollout

Table 6: Interval-rule comparisons with a margin of 0.05 on the normalized scale. A1 uses the same ES contrast as A2.

ID	Comparison	Difference [interval]	Decision
H1	OE minus PPO-DR	-0.5592 $[-0.6088, -0.5196]$	Against predicted superiority
A1	ES-DR minus ES-fixed	$+0.0286$ $[0.0203, 0.0384]$	Positive, below margin

length from $\{32, 64, 128\}$, update epochs from $\{2, 4, 8\}$, discount from $\{0.95, 0.99, 0.995\}$, entropy coefficient from $\{0, 0.001, 0.01\}$ and reward normalization on or off. ES sampled perturbation scale from $[0.01, 0.1]$ and learning rate from $[0.003, 0.03]$, both log-uniformly, population size from $\{32, 64, 128\}$ and episodes per candidate from $\{1, 2, 4\}$. OE sampled weight-mutation standard deviation log-uniformly from $[0.005, 0.05]$. Population floor and change interval were design choices, not tuned parameters. Each trial had its own tuning seed; a score tie favored the lower trial index.

Validation environments used seeds 4900–4919 plus four corner cases. Final T1, T2 and T3 environments used seeds 5900–5909, 5910–5919 and 5920–5929. Competence episodes were keyed from environment-specific evaluation seeds and checkpoint indices; the same test conditions were used across methods. Final run seeds were 5000–5009. The implementation separates named random streams by purpose, although common seeds can still couple methods. The exact realized environment parameters, rather than only their generating ranges, are retained with every evaluation record.

B Complete neural comparisons

H1 compares OE and PPO-DR on adaptation AUC (Equation 1); A1 compares ES-DR and ES-fixed. H1 lies below -0.05, while A1 is positive and below 0.05 (Table 6). The margin is defined on the empirical normalized scale. AUC includes starting competence and therefore measures performance during adaptation rather than adaptation speed.

The H1/A1 rule checks the interval in this order. An interval wholly above zero supports the predicted direction, qualified as below the margin if it also lies below $+0.05$. An interval wholly below zero is against that direction, qualified as below the margin if it also lies above -0.05. If neither condition holds, an interval contained in $[-0.05, +0.05]$ yields the protocol's no-meaningful-difference label; any other interval is inconclusive. Thus directional support is not the same as exceeding the practical margin.

Table 7: Complete core S1–S7 and later A2–A3 families. Differences are pooled IQMs except S7, which uses IQMs of per-run extinction fractions. Intervals are nominal 95% whole-run intervals. Holm correction is separate for the two families. The rank-test p values apply to per-run summaries, not directly to the IQM effects.

ID	Comparison	Outcome	Difference [interval]	Adjusted p
S1	OE minus ES-DR	T2/T3 AUC	-0.5871 $[-0.6364, -0.5475]$	0.0011
S2	OE minus OE-fixed	T2/T3 AUC	-0.1210 $[-0.1799, -0.0680]$	0.0044
S3	PPO-DR minus PPO-fixed	T2/T3 AUC	$+0.0099$ $[-0.0004, 0.0201]$	0.6232
S4	OE minus PPO-DR	T2/T3 zero-shot	-0.4648 $[-0.5570, -0.4073]$	0.0011
S5	OE minus PPO-DR	T2/T3 final	-0.6035 $[-0.6486, -0.5643]$	0.0011
S6	OE minus PPO-DR	T1 AUC	-0.5371 $[-0.5951, -0.4799]$	0.0011
S7	OE minus PPO-DR	T2/T3 extinction	$+0.3750$ $[0.2500, 0.4833]$	0.0011
A2	ES-DR minus ES-fixed*	T2/T3 AUC	$+0.0286$ $[0.0203, 0.0384]$	3.65×10^{-4}
A3	ES-DR minus ES-fixed*	T1 zero-shot	$+0.0645$ $[0.0499, 0.0788]$	3.65×10^{-4}

S1, S2, S5 and S7 are against their predicted OE directions. S3 does not support its predicted PPO direction. S4 and S6 were nondirectional and favor PPO-DR. A2 and A3 support the predicted ES-DR direction under the separate addendum family. The Mann–Whitney probability summary is $U/(n_X n_Y)$, with ties contributing one half; the test is not generally a test of arithmetic mean equality. S7 first reduces each run to its extinction fraction across T2/T3, then compares the distributions of those fractions.

For H2b, write $q_{ov}, q_{of}, q_{pv}, q_{pf}$ for pooled AUC IQMs of OE, OE-fixed, PPO-DR and PPO-fixed. The prespecified summaries are

$$V = \tfrac{1}{2}[(q_{ov} - q_{of}) + (q_{pv} - q_{pf})], \tag{3}$$

$$M = \tfrac{1}{2}[(q_{ov} - q_{pv}) + (q_{of} - q_{pf})], \tag{4}$$

$$I = (q_{ov} - q_{of}) - (q_{pv} - q_{pf}), \qquad \Delta = |V| - |M|. \tag{5}$$

Table 8 reports these contrasts among the four tuned configurations. Their selected hyperparameters and training conditions differ, so the contrasts do not isolate environmental variation. The interaction describes how the varied-minus-fixed difference changes between OE and PPO. Absolute-value contrasts are unstable when a component is close to zero.

A4 compares the ES varied-minus-fixed contrast with the corresponding OE contrast. Its difference is 0.149628, interval $[0.095763, 0.209123]$. A4 is descriptive: the OE outcome was known before the ES-fixed extension. Its component intervals come from the four-method resampling routine and need not exactly equal the intervals from separate two-method routines.

Table 8: Four-method contrasts. H2b covers the core configurations; A4 adds ES-fixed after the OE comparison was known.

Analysis	Quantity	Estimate	Nominal 95% interval
H2b	$\Delta = \|V\| - \|M\|$	-0.438162	[-0.474258, -0.405598]
H2b	I	-0.130959	[-0.190704, -0.076383]
H2b	M	-0.493721	[-0.524014, -0.467100]
H2b	V	-0.055559	[-0.085238, -0.028180]
A4	I	+0.149628	[0.095763, 0.209123]
A4	ES-DR minus ES-fixed	+0.028590	[0.020276, 0.038461]
A4	OE minus OE-fixed	-0.121038	[-0.179733, -0.067839]

Table 9: Mean normalized performance, averaging environments within runs and then runs. Asterisk: ES-fixed extension.

Procedure	Suite	Zero-shot	Final	AUC
OE	T1	0.4052	0.3946	0.3960
OE	T2	0.4383	0.2338	0.3205
OE	T3	0.2966	0.2655	0.2759
OE-fixed	T1	0.5923	0.5088	0.5303
OE-fixed	T2	0.5285	0.2897	0.3860
OE-fixed	T3	0.5441	0.4076	0.4510
PPO-DR	T1	0.9236	0.9235	0.9295
PPO-DR	T2	0.7244	0.7478	0.7611
PPO-DR	T3	0.8981	0.8840	0.8861
PPO-fixed	T1	0.9161	0.9196	0.9216
PPO-fixed	T2	0.7181	0.7123	0.7290
PPO-fixed	T3	0.9202	0.9147	0.9089
ES-DR	T1	0.9017	0.9270	0.9231
ES-DR	T2	0.7114	0.8833	0.8743
ES-DR	T3	0.8543	0.8772	0.8686
ES-fixed*	T1	0.8223	0.9067	0.8930
ES-fixed*	T2	0.6770	0.8656	0.8377
ES-fixed*	T3	0.8041	0.8530	0.8470

C Sensitivity analyses and descriptive diagnostics

Post hoc re-estimation of the core maximum anchor inside each of 50,000 bootstrap draws gives intervals of $[-0.61045, -0.52146]$ for H1, $[-0.17997, -0.06776]$ for S2, $[0.02042, 0.03851]$ for A1 and $[0.04965, 0.07903]$ for A3. Paired resampling of corresponding seeds gives $[-0.61012, -0.51847]$, $[-0.18100, -0.06919]$ and $[0.02039, 0.03964]$ for H1, S2 and A1. For S3, the paired IQM interval is $[0.0020, 0.0177]$, above zero, although the prespecified rank test on run means does not support the predicted PPO-DR advantage. This sensitivity uses both a different performance summary and a different treatment of cross-method dependence.

Realized adaptation counts differ between methods. The mean final counts per environment are

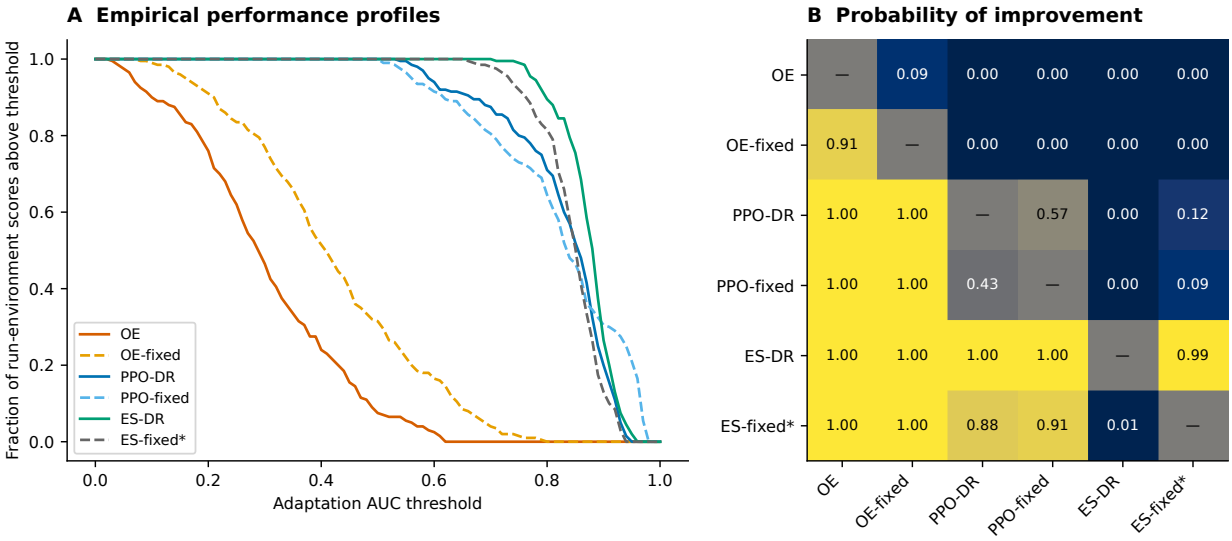


Figure 4: Descriptive performance summaries. Left: fraction above each AUC threshold among ten runs across twenty fixed environments. Right: fraction of the 100 cross-run comparisons in which the row method's mean exceeds the column method's mean, with ties counting one half.

approximately 20,000,010 for either OE variant, 20,017,188 for PPO-DR, 20,039,595 for PPO-fixed, 20,144,256 for ES-DR and 20,288,091 for ES-fixed. ES-fixed can overshoot its first two-million-step checkpoint by 49.5%; the largest checkpoint excess is 4.98% of the complete adaptation budget, and the largest final excess is 4.49%. Actual-step sensitivity integrates the piecewise-linear saved observations and truncates at 20 million steps. Scores between saved checkpoints are linearly interpolated. The resulting H1 estimate is -0.559203; the ES contrast is 0.028921, with interval $[0.020572, 0.038833]$.

The primary deficit persists when individual runs or environments are omitted and when raw scores replace normalized scores. Figure 4 shows empirical performance profiles and pairwise run-mean probability summaries.

The time-to-threshold diagnostic compares raw competence against 90% of each run's mean T1 zero-shot score. In the core study, OE and OE-fixed each reach that threshold in 43 of 200 T2/T3 run–environment pairs; PPO-DR, PPO-fixed and ES-DR each reach it in 40. The threshold depends on the source-environment score and may be unattainable in the target environment.

Weight diversity is mean pairwise Euclidean distance among sixteen sampled neural parameter vectors; behavior diversity uses action distances on 256 fixed validation probes. Both measures are higher for OE than OE-fixed at the final checkpoint (Figure 5). Body-trait diversity is zero by design.

Three exploratory OE runs extend training to one billion steps (Figure 6). Their gains are -4.3657, 1.8893 and -0.9751, measured as the mean of the final two validation checkpoints minus the mean of the first two. These results are descriptive; the planned 4.3-point threshold used an unreliable noise estimate.

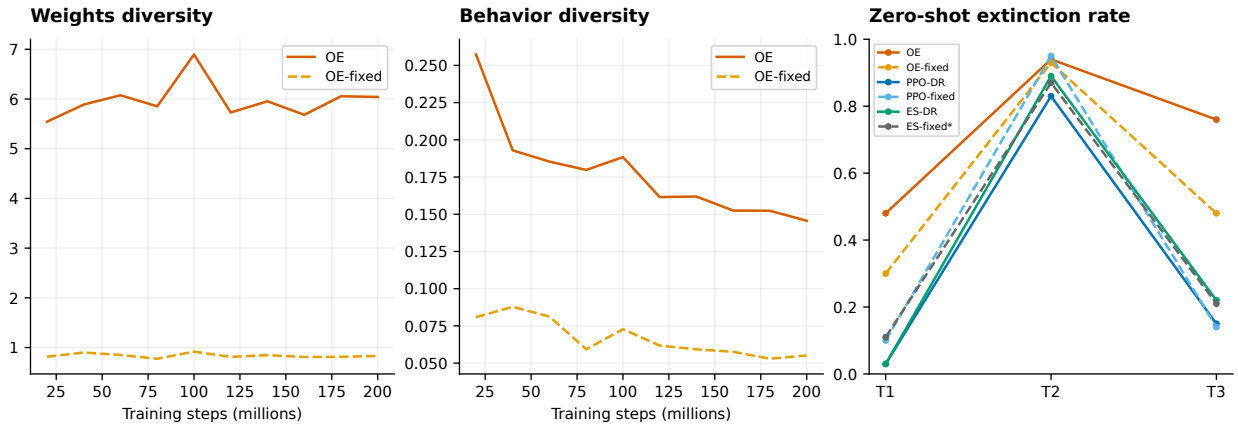


Figure 5: Population diagnostics and separate persistence. Weight and behavioral diversity summarize sixteen sampled agents per training population, averaged across ten runs. Right: extinction frequencies in the separate persistence assay without floor support.

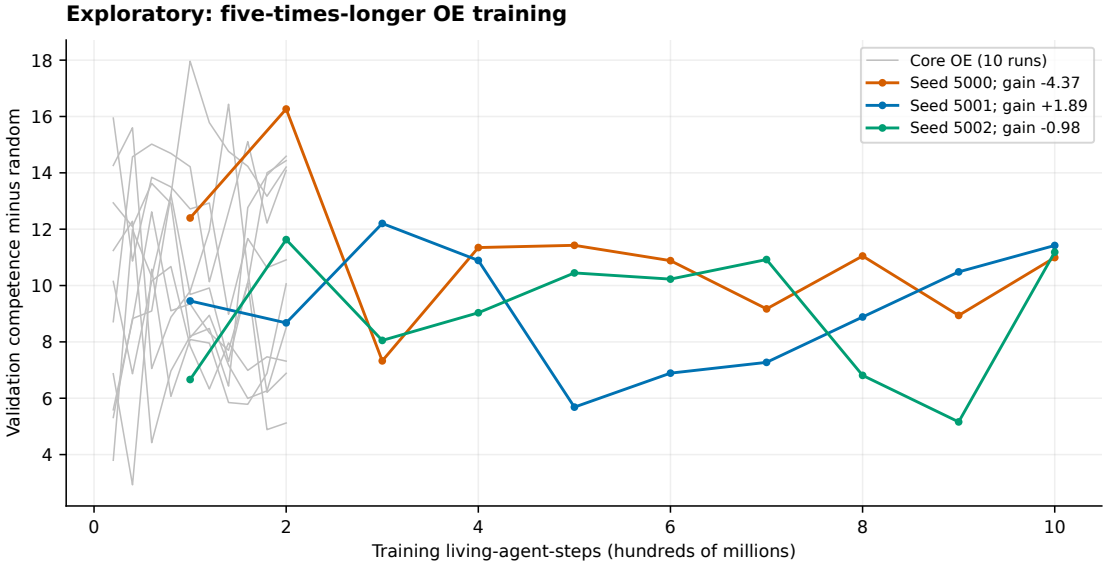


Figure 6: Exploratory longer training. Three fresh OE runs trained to one billion nominal steps, with the ten original 200-million-step runs in gray. Gain is the mean of the final two saved checkpoints minus the mean of the first two.

D Supporting trait-only experiment

D.1 Design and analysis

The trait model combines inherited speed, size, sensing range and reach with fixed behavioral rules. Populations spend 100 generations in environment A and 50 in B, except Oracle-B, which starts directly in B. Each final run seed defines its own searches and deployments. There are 200 seeds per primary group and arm (2000–2199). Shared food, behavior and initial-position streams couple groups with the same seed. The main B has 25 food items; three additional arms permit agent consumption in B, enlarge the B map to side 750, or reduce B's energy budget per generation to 25. The larger-map and reduced-energy settings were selected during recorded pilot and amendment rounds before the corresponding final runs.

The generation rules derive from the MinuteLabs Evolution Simulator at commit `55af956` (Palfree, 2021). A has side length 500 and 50 uniformly placed food items. Main B has side length 300 and 25 items in five clusters with standard deviation 15; centers and food are redrawn each generation. A generation lasts at most 200 ticks, with an energy allowance of 500 per agent except in the reduced-energy B arm. Agents start at edge home positions, search using fixed steering rules and return home according to feeding and remaining energy. Exhaustion and, in the consumption arm, predation can kill agents; agents with no food at generation end also die. Survivors carry into the next generation, and each survivor that consumed at least two food units produces exactly one offspring. Food units count both food items and prey. Effective movement speed is $u = \text{speed} \times \text{size}/10$, and each active movement tick costs $10^{-4}(\text{size}^3 u^2 + \text{sense})$. Food supports survival and reproduction without replenishing this generation-level energy allowance. There is no externally supported population floor.

Initial populations have 50 agents. ET starts from speed/size/sense/reach values 10/10/20/1. Mutating offspring receive independent Gaussian trait perturbations with standard deviations 1/1/1/0.25; sensing range is clipped at zero and the other traits at the smallest positive normalized float64 value. ET and M-pop mutate in both phases. F-food, F-pop, SV and Oracle-B have mutation disabled throughout. ET-freeze and ET-clone are derived from ET after A and have mutation disabled in B. SV samples each trait independently and log-uniformly over the search ranges: speed and size 2.5–40, sense 5–80 and raw reach 0.25–4.

In the consumption arm, a predator can consume only an agent whose size is at most 80% of its own. Identical clones therefore cannot consume one another. Comparisons involving clones and heterogeneous populations also differ in access to this additional food source; differences among heterogeneous groups depend on their size distributions. Across all 50 B generations, mean prey counts per run are 150.8 for ET, 153.2 for M-pop, 18.4 for ET-freeze and 3.4 for SV, with zero for the four homogeneous frozen groups.

The A search begins with eight logarithmically spaced values for each trait. Merging designs with the same effective reach, $\max(\text{reach}, \text{size}/4)$, leaves 1,472 distinct designs. Stage one evaluates 50 clones of each design for 30 generations on shared streams. Stage two evaluates the top 32 designs per objective on eight fresh sets of streams. In stage two, population search scores mean population over generations 10–99; food search scores food per agent over generations 10–29, even when a candidate was simulated for longer. The comparison therefore changes both objective and scoring horizon. Search-selected mutating agents inherit effective reach.

The group identifiers used in the tables are: ET, the evolving population; F-food and F-pop, frozen copies of the food- and population-search winners; M-pop, mutating copies of the population-search winner; SV, a frozen initial population with standing variation; ET-freeze, the end-of-A

evolved population with mutation disabled in B; ET-clone, copies of a randomly sampled end-of-A member with mutation disabled; and Oracle-B, the reference design selected directly in B. Oracle-B is not part of the confirmatory contrast family. Cloning changes the realized traits along with heterogeneity.

P1 is population at the start of a generation, averaged over B generations 40–49 with zeros retained. P2 indicates that no agent would start B generation 50. Each arm has eight planned comparison pairs, except arm 2 omits the population-search versus food-search pair because those data repeat the main arm by construction: 31 pairs in total. P1 uses a two-sided 10,000-permutation test of mean differences and a studentized bootstrap interval with 20,000 draws. P2 uses a two-sided Fisher test and a Newcombe interval for the difference in extinction proportions. Holm adjustment is within each arm's 16-test family, or 14 tests for arm 2. Shared streams make independent-group test calibration an approximation; paired sensitivity is retained separately in the supporting analysis records.

The practical margins are 2.5 agents for P1 and 0.10 for P2. The original "meaningfully different" label requires adjusted $p < 0.05$ and an absolute point estimate at least as large as the margin; it does not require the entire interval to exceed it. Equivalence uses an interval at level $1 - 2(0.05)/k$, where k is the arm's family size, wholly inside the margins. Boundary labels identify ceiling and floor effects that can mask population differences.

D.2 Results

In the main arm, F-pop reaches the population ceiling while ET is close to it. Their mean population difference is 0.6375 against a 2.5-agent margin, and neither group becomes extinct. F-pop also outperforms F-food, but their searches differ in both objective and scoring horizon. Cloning reduces population relative to the frozen diverse population after the move; a larger deficit occurs in the no-move control. Table 10 reports all 31 pairs and both outcomes. Table 11 gives every primary group's mean population, extinction count, pre-shift losses and boundary frequencies. Full labels are in `data/trait_complete_contrasts.csv`; Figure 7 shows group outcomes.

Table 10: All 31 trait-study comparison pairs; all differences are X minus Y, with comparisons written X / Y. P1 is mean late-B population and P2 is terminal extinction probability. Entries are differences with 95% intervals. MD: significant with point estimate at or beyond the margin; EQ: equivalence under the specified interval rule; DU: detected difference with size unresolved; IN: inconclusive. Boundary qualifications are in Table 11; full labels are retained in the accompanying CSV.

Arm-row	X / Y	P1 difference [CI]	P2 difference [CI]	Adj. p_1 / p_2	P1/P2
main-1	F-food / ET	-20.93 $[-21.66, -20.11]$	$+0.71$ $[0.64, 0.77]$	0.0013 / 9.54×10^{-60}	MD/MD
main-2	F-pop / ET	$+0.64$ $[0.59, 0.68]$	$+0.00$ $[-0.02, 0.02]$	0.0013 / 1	EQ/EQ
main-3	M-pop / ET	-0.02 $[-0.08, 0.04]$	$+0.00$ $[-0.02, 0.02]$	1 / 1	EQ/EQ
main-4	SV / ET	-0.16 $[-0.64, 0.10]$	$+0.00$ $[-0.02, 0.02]$	1 / 1	EQ/EQ
main-5	ET-freeze / ET	-0.23 $[-0.41, -0.08]$	$+0.00$ $[-0.02, 0.02]$	0.04 / 1	EQ/EQ
main-6	ET-clone / ET-freeze	-3.11 $[-4.36, -2.11]$	$+0.10$ $[0.06, 0.15]$	0.0013 / 1.62×10^{-5}	MD/MD

Arm-row	X / Y	P1 difference [CI]	P2 difference [CI]	Adj. p_1 / p_2	P1/P2
main-7	F-pop / F-food	$+21.57$ $[20.76, 22.28]$	-0.71 $[-0.77, -0.64]$	0.0013 / 9.54×10^{-60}	MD/MD
main-8	M-pop / F-pop	-0.66 $[-0.70, -0.62]$	$+0.00$ $[-0.02, 0.02]$	0.0013 / 1	EQ/EQ
A2-1	F-food / ET	-19.16 $[-20.02, -18.17]$	$+0.70$ $[0.63, 0.76]$	0.0012 / 5.18×10^{-56}	MD/MD
A2-2	F-pop / ET	$+2.41$ $[1.97, 3.02]$	-0.01 $[-0.04, 0.01]$	0.0012 / 1	DU/EQ
A2-3	M-pop / ET	$+1.68$ $[1.05, 2.34]$	-0.01 $[-0.03, 0.02]$	0.0012 / 1	DU/EQ
A2-4	SV / ET	$+1.40$ $[0.75, 2.05]$	-0.01 $[-0.03, 0.02]$	0.0012 / 1	EQ/EQ
A2-5	ET-freeze / ET	$+1.24$ $[0.76, 1.84]$	-0.01 $[-0.04, 0.01]$	0.0012 / 1	EQ/EQ
A2-6	ET-clone / ET-freeze	-2.80 $[-4.03, -1.81]$	$+0.10$ $[0.06, 0.15]$	0.0012 / 1.50×10^{-5}	MD/MD
A2-8	M-pop / F-pop	-0.73 $[-1.26, -0.38]$	$+0.01$ $[-0.01, 0.03]$	0.0018 / 1	EQ/EQ
A3-1	F-food / ET	-18.10 $[-18.83, -17.18]$	$+0.94$ $[0.89, 0.96]$	0.0012 / 5.98×10^{-98}	MD/MD
A3-2	F-pop / ET	$+5.74$ $[5.00, 6.66]$	-0.07 $[-0.11, -0.03]$	0.0012 / 0.0012	MD/DU
A3-3	M-pop / ET	$+5.53$ $[4.80, 6.43]$	-0.07 $[-0.11, -0.03]$	0.0012 / 0.0012	MD/DU
A3-4	SV / ET	-6.86 $[-8.56, -5.13]$	$+0.39$ $[0.31, 0.46]$	0.0012 / 1.48×10^{-18}	MD/MD
A3-5	ET-freeze / ET	-0.50 $[-1.87, 0.81]$	$+0.07$ $[0.02, 0.14]$	0.9333 / 0.07	EQ/IN
A3-6	ET-clone / ET-freeze	-14.16 $[-15.54, -12.58]$	$+0.66$ $[0.58, 0.72]$	0.0012 / 1.44×10^{-41}	MD/MD
A3-7	F-pop / F-food	$+23.84$ $[23.70, 23.97]$	-1.00 $[-1.00, -0.97]$	0.0012 / 3.11×10^{-118}	MD/MD
A3-8	M-pop / F-pop	-0.22 $[-0.39, -0.04]$	$+0.00$ $[-0.02, 0.02]$	0.07 / 1	EQ/EQ
A4-1	F-food / ET	-13.43 $[-15.02, -11.81]$	$+0.57$ $[0.50, 0.64]$	0.0012 / 3.99×10^{-44}	MD/MD
A4-2	F-pop / ET	$+0.12$ $[-1.98, 2.27]$	-0.14 $[-0.23, -0.04]$	1 / 0.0329	IN/MD
A4-3	M-pop / ET	-0.76 $[-3.04, 1.49]$	$+0.04$ $[-0.06, 0.13]$	1 / 1	IN/IN
A4-4	SV / ET	-10.58 $[-12.43, -8.63]$	$+0.45$ $[0.36, 0.53]$	0.0012 / 1.01×10^{-20}	MD/MD
A4-5	ET-freeze / ET	-3.93 $[-6.12, -1.66]$	$+0.14$ $[0.05, 0.24]$	0.0032 / 0.0303	MD/MD
A4-6	ET-clone / ET-freeze	-8.97 $[-10.63, -7.33]$	$+0.41$ $[0.33, 0.48]$	0.0012 / 1.39×10^{-23}	MD/MD
A4-7	F-pop / F-food	$+13.55$ $[12.15, 14.93]$	-0.71 $[-0.77, -0.64]$	0.0012 / 9.54×10^{-60}	MD/MD
A4-8	M-pop / F-pop	-0.88 $[-3.02, 1.24]$	$+0.17$ $[0.08, 0.26]$	1 / 0.0044	IN/MD

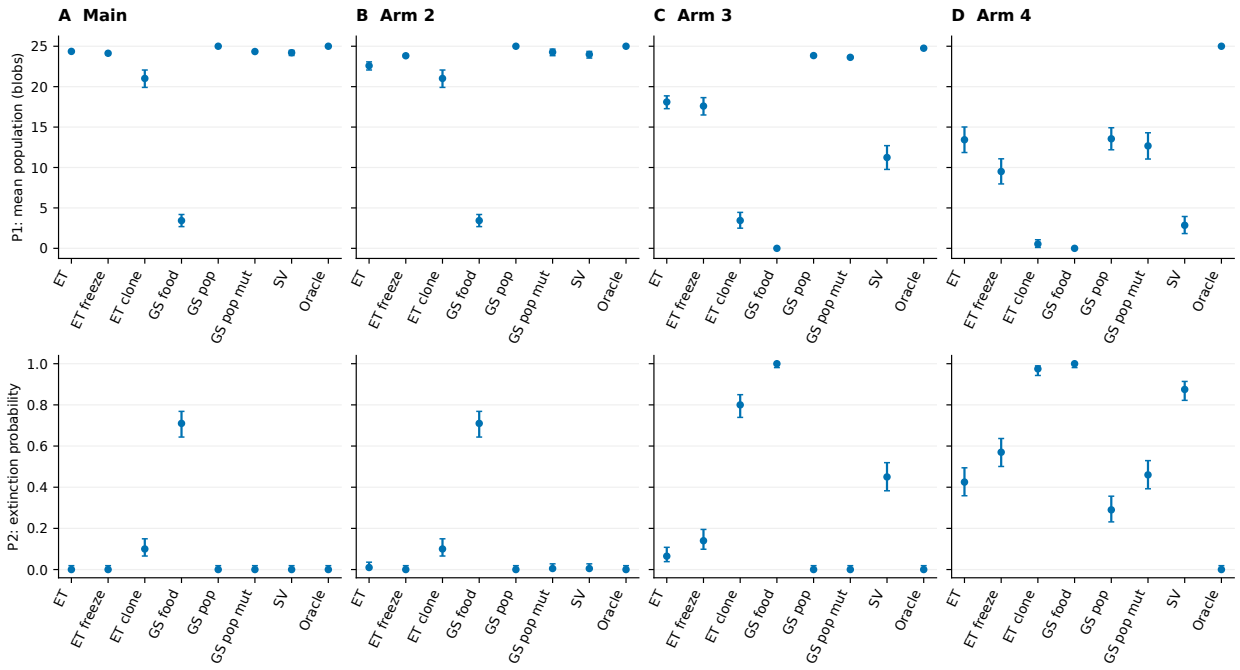


Figure 7: Supporting trait experiment. All eight primary groups in four arms, with 200 seeds per group. Upper panels: mean P1 and descriptive whole-seed percentile 95% intervals (20,000 draws). Lower panels: P2 proportions with Wilson 95% intervals. These differ from the studentized P1 and Newcombe P2 intervals for contrasts.

Table 11: Trait-study group outcomes and boundary diagnostics. All groups have 200 seeds. High/low are fractions with P1 at least 22.5 / at most 2.5. In arm 2, 25 is not a hard population ceiling. A-dead identifies losses preceding the environmental shift.

Arm	Group	Mean P1	Extinct	A-dead	High	Low
main	F-food	3.43	142/200	62	0.00	0.69
main	ET	24.36	0/200	0	1.00	0.00
main	F-pop	25.00	0/200	0	1.00	0.00
main	M-pop	24.34	0/200	0	1.00	0.00
main	SV	24.20	0/200	0	0.92	0.00
main	ET-freeze	24.13	0/200	0	0.90	0.00
main	ET-clone	21.02	20/200	0	0.73	0.10
main	Oracle-B	25.00	0/200	0	1.00	0.00
A2	F-food	3.43	142/200	62	0.00	0.69
A2	ET	22.59	2/200	0	0.61	0.01
A2	F-pop	25.00	0/200	0	1.00	0.00
A2	M-pop	24.27	1/200	0	0.85	0.01
A2	SV	23.99	1/200	0	0.90	0.01
A2	ET-freeze	23.82	0/200	0	0.86	0.00
A2	ET-clone	21.02	20/200	0	0.73	0.10
A2	Oracle-B	25.00	0/200	0	1.00	0.00
A3	F-food	0.00	200/200	62	0.00	1.00
A3	ET	18.10	13/200	0	0.10	0.07

Arm	Group	Mean P1	Extinct	A-dead	High	Low
A3	F-pop	23.84	0/200	0	0.93	0.00
A3	M-pop	23.63	0/200	0	0.94	0.00
A3	SV	11.24	90/200	0	0.24	0.45
A3	ET-freeze	17.60	28/200	0	0.33	0.14
A3	ET-clone	3.44	160/200	0	0.04	0.79
A3	Oracle-B	24.76	0/200	0	1.00	0.00
A4	F-food	0.00	200/200	62	0.00	1.00
A4	ET	13.43	85/200	0	0.50	0.42
A4	F-pop	13.55	58/200	0	0.34	0.28
A4	M-pop	12.67	92/200	0	0.54	0.46
A4	SV	2.85	175/200	0	0.09	0.88
A4	ET-freeze	9.50	114/200	0	0.28	0.56
A4	ET-clone	0.54	195/200	0	0.01	0.97
A4	Oracle-B	25.00	0/200	0	1.00	0.00

In the main arm, the food-search group's deficit includes 62 of 200 runs lost during A. Restricting analysis to groups that both reach B selects survivors and can bias the comparison; such conditional outcomes are retained as diagnostics rather than replacing unconditional results. The no-move control continues ET's end-of-A population for fifty further generations in A. Mean late population is 48.0675 with mutation, 47.9895 after freezing, and 40.9730 after cloning. The freeze-minus-mutation difference is -0.0780, interval $[-0.2720, 0.1140]$, which includes zero. The clone-minus-freeze difference is -7.0165, interval $[-9.1271, -5.0470]$. The clone deficit therefore also occurs without an environmental shift.

Searching directly in B improves population score by less than 2.5 agents over searching in A in 200 of 200 main-arm runs, 200 of 200 consumption-arm runs, 199 of 200 larger-map runs and 67 of 200 reduced-energy runs. Thus searching directly in B rarely improves population score by at least 2.5 agents in the first three settings. The standing-variation group retains only one design at transfer in 153 of 200 runs, limiting its use as a diverse-population control.

Additional exploratory results and sample sizes are in `data/trait_exploratory.json`. On a 100-seed subset, starting ET directly in B instead of transferring from A changes P1 by -0.22 in the main arm, -17.62 in the larger-map arm and -12.85 in the reduced-energy arm.

E Protocols and reproducibility

The core protocols were written before the corresponding runs and were not publicly registered. ES-fixed was specified after inspection of the core results, before its own tuning and final runs. The one-billion-step OE extension was exploratory. A three-seed pilot informed the choice of ten final runs per configuration, which limits precision for equivalence tests.

Neural training and evaluation records identify revision `16e8657` for the core experiment and `06ff7b2` for ES-fixed. Trait generation records identify `a1e1cdc`; its confirmatory analysis identifies `561bcd4`. Full revision identifiers and configuration hashes accompany the code.

Core neural runs used NVIDIA RTX 4090 hardware, JAX 0.11.2 and Python 3.13.18. Statistical reanalysis used Python 3.12.14 on CPU.

The data include all sixty final neural evaluations, 19,800 adaptation checkpoint observations, trait generation records, final policies, training snapshots and tuning records. Core tuning policy files were not retained. The repository's reproduction guide gives commands for rebuilding the tables, figures and statistical analyses.